

# Self-Study Notes: A Coherent Asset Pricing Roadmap via the Stochastic Discount Factor

## 0. Orientation

- **Central organizing object:** stochastic discount factor (SDF)  $m_{t+1}$ .
- **One pricing identity:** all payoffs (stocks, bonds, derivatives) satisfy the same pricing restriction.
- **Main coherence rule:** each chapter either (i) derives  $m_{t+1}$ , (ii) characterizes admissible  $m_{t+1}$ , (iii) uses  $m_{t+1}$  to price assets, (iv) diagnoses why simple  $m_{t+1}$  fail, or (v) estimates/tests  $m_{t+1}$  in data.

## Notation

- $p_t$ : time- $t$  price;  $x_{t+1}$ : time- $t+1$  payoff;  $R_{t+1} = x_{t+1}/p_t$ : gross return.
- $R_{f,t}$ : gross risk-free rate;  $R_{i,t+1}^e = R_{i,t+1} - R_{f,t}$ : excess return.
- $\mathbb{E}_t[\cdot]$ : conditional expectation given information at  $t$ ;  $\text{Cov}_t(\cdot, \cdot)$ : conditional covariance.

## 1. The pricing spine: SDF and its immediate consequences

### 1.1 Master equation

$$p_t = \mathbb{E}_t[m_{t+1} x_{t+1}], \quad \Longleftrightarrow \quad 1 = \mathbb{E}_t[m_{t+1} R_{t+1}]. \quad (1)$$

### 1.2 Risk premia as covariance with the SDF

For any excess return  $R_{i,t+1}^e$ :

$$0 = \mathbb{E}_t[m_{t+1} R_{i,t+1}^e] \quad \Rightarrow \quad \mathbb{E}_t[R_{i,t+1}^e] = -\frac{\text{Cov}_t(m_{t+1}, R_{i,t+1}^e)}{\mathbb{E}_t[m_{t+1}]}.$$
 (2)

Interpretation: assets earn premia when they pay off poorly in states where  $m_{t+1}$  is high.

### 1.3 Risk-neutral measure

Define a probability measure  $Q$  via the Radon–Nikodym derivative proportional to  $m_{t+1}$ :

$$m_{t+1} = \frac{1}{R_{f,t}} \frac{dQ}{dP} \Big|_{t+1}. \quad (3)$$

Then pricing becomes:

$$p_t = \frac{1}{R_{f,t}} \mathbb{E}_t^Q[x_{t+1}], \quad (4)$$

which is the bridge to term structure and derivative pricing.

### 1.4 Linear factor models as parametric SDFs

A linear factor structure posits:

$$m_{t+1} = a_t - b_t^\top f_{t+1}. \quad (5)$$

Plugging (5) into  $\mathbb{E}_t[m_{t+1}R_{i,t+1}^e] = 0$  yields beta pricing restrictions (APT/CAPM/ICAPM families).

### 1.5 Hansen–Jagannathan (HJ) bound as a diagnostic

A common diagnostic inequality:

$$\frac{\sigma_t(m_{t+1})}{\mathbb{E}_t[m_{t+1}]} \geq \max_i \frac{|\mathbb{E}_t[R_{i,t+1}^e]|}{\sigma_t(R_{i,t+1}^e)}. \quad (6)$$

Interpretation: observed Sharpe ratios imply a minimum volatility for admissible  $m_{t+1}$ .

### 1.6 Present value decomposition link

Predictability topics can be organized as: variation in prices reflects either expected cash flows or expected discount rates. Campbell–Shiller style logic operationalizes this separation.

## 2. Coherence map: how the parts connect

### 2.1 Narrative chain

1. **Preferences  $\rightarrow$  Euler equation  $\rightarrow$  SDF (Part I, Ch. 1).**
2. **Mean–variance special cases (Part I, Ch. 2):** connects portfolio geometry to pricing restrictions.
3. **Complete markets equilibrium (Part I, Ch. 3):** state prices and risk-neutral measure emerge cleanly.
4. **Incomplete markets no-arbitrage (Part I, Ch. 4):** existence of SDF without uniqueness.
5. **SDF diagnostics (Part I, Ch. 5):** bounds/conditioning discipline candidate SDFs.

6. **Factor models (Part I, Ch. 6):** structured families of SDFs with testable implications.
7. **Predictability and valuation (Part I, Ch. 7):** discount-rate dynamics become measurable objects.
8. **Puzzles (Part II, Ch. 8):** baseline SDFs fail to match key moments/bounds.
9. **Solutions (Part II, Ch. 9–12):** “SDF amplifiers” modify volatility/state dependence/tail behavior of  $m_{t+1}$ .
10. **Derivatives (Part III, Ch. 13–14):** replication and risk-neutral pricing are applications of (4).
11. **Advanced topics (Part IV):** enrich state dynamics, preferences, model uncertainty, and environments while preserving (1).
12. **Empirics (Part V):** estimate/test  $m_{t+1}$  or its factor representation; close the theory–data loop.

## 2.2 One-page visual map

### 3. Part I notes: Basic theoretical framework (core toolkit)

#### Ch. 1 — Consumption-based theory

**Goal:** derive  $m_{t+1}$  from preferences and obtain (2).

- **Key output:** Euler equation  $\Rightarrow$  explicit SDF (e.g., basic CRRA implies  $m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}$ ).
- **Core implications:** risk correction, risk premium decomposition, risk-neutral pricing via (4).
- **Common friction points:** equity premium and risk-free rate behavior; continuous-time vs discrete-time mapping.

#### Ch. 2 — Static portfolio theory (mean–variance and CAPM)

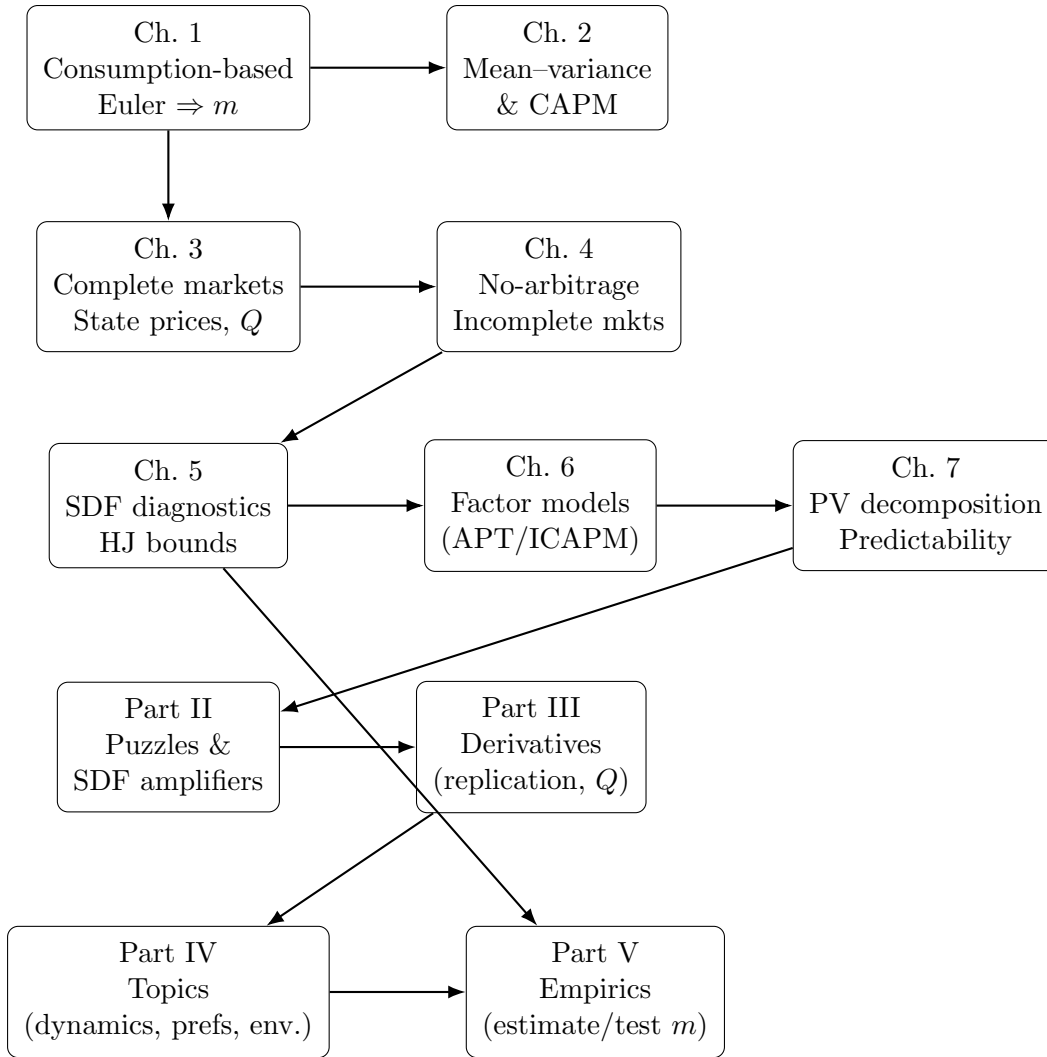
**Goal:** connect return geometry to pricing restrictions.

- Efficient frontier; tangency portfolio; two-fund separation.
- CAPM equilibrium restrictions; interpretation as a single-factor representation.
- Bridge to SDF: mean–variance sufficiency corresponds to an SDF that is (approximately) affine in returns.

#### Ch. 3 — General equilibrium (complete markets)

**Goal:** state prices and pricing kernel in equilibrium.

- State prices  $\pi(s)$  and pricing kernel viewpoint.
- Risk-neutral pricing emerges as a change of measure induced by  $m_{t+1}$ .
- Aggregation/risk sharing: representative-agent mapping is tractable under completeness.



## Ch. 4 — No-arbitrage (incomplete markets)

**Goal:** pricing without completeness; existence of (possibly many) SDFs.

- Law of one price / no-arbitrage  $\Rightarrow$  existence of at least one  $m_{t+1}$  satisfying (1).
- Non-uniqueness outside the traded payoff space; projection arguments within payoff space.
- Fundamental theorem of finance: relation between no-arbitrage and equivalent martingale measures.

## Ch. 5 — Stochastic discount factor (diagnostics and bounds)

**Goal:** discipline  $m_{t+1}$  via volatility bounds and conditional restrictions.

- HJ bounds (6); minimum-variance SDF; interpretation via Sharpe ratios.
- Conditional SDF: information sets and conditional moment restrictions.
- Translation to mean-variance: equivalence links to Hansen–Richard style results.

## Ch. 6 — Linear factor models (APT/CAPM/ICAPM)

**Goal:** impose structure on  $m_{t+1}$  to obtain testable beta pricing.

- APT exact vs approximate factor pricing; relationship to traded vs non-traded factors.
- CAPM as special case; ICAPM adds time-varying investment opportunities and hedging demand.
- Conditional vs unconditional beta pricing; managed portfolios as conditional instruments.

## Ch. 7 — Present value decomposition (predictability)

**Goal:** interpret predictability through discount-rate variation vs cash-flow variation.

- Campbell–Shiller decomposition; predictive regressions; small-sample bias concerns.
- Connection to SDF: time variation in expected returns corresponds to state dependence in  $m_{t+1}$ .

## 4. Part II notes: Puzzles and “SDF amplifiers”

### Ch. 8 — The dual puzzle

**Core diagnosis:** baseline consumption-based  $m_{t+1}$  often lacks sufficient volatility/state dependence to satisfy (6) while matching the risk-free rate and equity premium.

- Euler-equation tests; equity premium puzzle; risk-free rate puzzle.
- Volatility bounds as a unifying diagnostic for multiple anomalies.

### Ch. 9–12 — Candidate mechanisms

Organizing principle: each mechanism modifies  $m_{t+1}$  by changing at least one of:

(i) volatility, (ii) state dependence, (iii) tail risk / rare events, (iv) effective consumption risk.

| Mechanism                          | How $m_{t+1}$ changes                                                        | Primary targets                                |
|------------------------------------|------------------------------------------------------------------------------|------------------------------------------------|
| Epstein–Zin / long-run risk        | Separates risk aversion and IES; amplifies long-run components               | Equity premium, term structure, predictability |
| Incomplete markets / heterogeneity | Idiosyncratic risk and limited sharing raise effective marginal utility risk | Equity premium, cross-sectional patterns       |
| Rare disasters                     | Adds tail states with large $m_{t+1}$ weight                                 | High premia with plausible consumption moments |
| Habit persistence                  | Time-varying risk aversion via surplus consumption                           | Countercyclical premia, volatility             |

## 5. Part III notes: Financial derivatives as SDF applications

### Ch. 13 — Brownian motion toolkit

- Stochastic integrals; Itô's lemma; martingales.
- Continuous-time language for (4): discounted prices are  $Q$ -martingales.

### Ch. 14 — Option pricing

- Replication/absence of arbitrage  $\Rightarrow$  Black–Scholes PDE.
- Risk-neutral derivation: (4) with diffusion dynamics.
- Extensions: puts, American exercise, stochastic volatility, stochastic rates.
- Unifying perspective: derivative pricing uses the same  $m_{t+1}$  (or the same  $Q$ ) as underlying assets.

## 6. Part IV notes: Advanced modules (same pricing spine, richer environments)

- **Martingale methods & dynamic portfolio choice:** solve intertemporal problems; clarify completeness; connect to hedging demands.
- **Learning under uncertainty:** beliefs as state variables; regime switching; Kalman filtering.
- **Markovian asset pricing & long-horizon valuation:** eigenfunction methods; long-run component extraction; shock elasticities.
- **Robustness to misspecification:** model uncertainty changes effective SDF (worst-case tilting).
- **Production economies:** investment/adjustment costs; link between real decisions and asset prices.
- **Demand-based asset pricing:** investor demand systems; exchange rates and global demand; investor types.
- **Recursive utility extensions:** continuous-time recursive utility; combinations with habit.

## 7. Part V notes: Reduced-form empirical analysis (closing the loop)

### Core empirical objects

- **SDF estimation:** recover  $m_{t+1}$  that best fits  $\mathbb{E}[mR^e] = 0$  across many assets.
- **Factor model testing:** estimate  $(\beta, \lambda)$  and test pricing errors with GMM.
- **Conditional pricing:** managed portfolios and instruments implement conditional moments.
- **High-dimensional predictors:** shrinkage, PCA/IPCA, and machine learning methods.

## Minimal empirical checklist

1. Choose test assets (portfolios/industries/bonds/options) and measurement frequency.
2. Choose model class (linear factor, conditional factor, SDF extraction).
3. Specify moments:  $\mathbb{E}[mR^e] = 0$  (unconditional) or  $\mathbb{E}[z_t m R^e] = 0$  (conditional with instruments  $z_t$ ).
4. Estimation method: OLS/Fama–MacBeth, GMM, PCA/IPCA, ML prediction of returns/discount rates.
5. Evaluate: pricing errors, Sharpe/HJ implications, out-of-sample performance, overfitting diagnostics.

## 8. Consistency template for each subsection

Each subsection can be written using the same five-line template:

1. **Object:** specify payoff/return process being priced.
2. **Assumptions:** equilibrium (preferences/consumption) vs no-arbitrage (trading restrictions/market structure).
3. **Restriction:** write  $1 = \mathbb{E}_t[m_{t+1}R_{t+1}]$  and characterize  $m_{t+1}$  (explicit, set-valued, or parametric).
4. **Implication:** risk premia, bounds, factor pricing, predictability, or derivative prices.
5. **Empirical hook:** moment conditions, estimation approach, testable pricing errors.

## 9. Study schedules (time plans)

### Plan A: Core spine (8 sessions $\times$ 2 hours)

| Session | Coverage              | Target competence                                                    |
|---------|-----------------------|----------------------------------------------------------------------|
| 1       | Ch. 1 (core)          | Derive Euler equation; interpret $m_{t+1}$ ; compute premia via (2). |
| 2       | Ch. 2                 | Relate mean–variance geometry to linear SDF intuition.               |
| 3       | Ch. 3                 | Work with state prices and risk-neutral mapping (3).                 |
| 4       | Ch. 4                 | State no-arbitrage SDF existence; interpret non-uniqueness.          |
| 5       | Ch. 5                 | Apply HJ bound (6); interpret minimum SDF volatility.                |
| 6       | Ch. 6                 | Translate factor models into SDF form (5); understand beta pricing.  |
| 7       | Ch. 7                 | Interpret predictability via discount-rate vs cash-flow news.        |
| 8       | Ch. 8 + one mechanism | Diagnose dual puzzle; map mechanism to changes in $m_{t+1}$ .        |

## Plan B: Full coherent arc (12 weeks)

| Week | Coverage                                  | Coherence target                                               |
|------|-------------------------------------------|----------------------------------------------------------------|
| 1    | Ch. 1 (two-period + multi-period)         | Build SDF and risk-premium logic.                              |
| 2    | Ch. 1 (portfolio choice) + Ch. 2          | Connect choice $\rightarrow$ mean-variance $\rightarrow$ CAPM. |
| 3    | Ch. 3 (complete markets)                  | State prices and $Q$ as shadow prices.                         |
| 4    | Ch. 4 (incomplete markets)                | Pricing without completeness; SDF set.                         |
| 5    | Ch. 5 (bounds, conditional SDF)           | Discipline candidate SDFs; conditional moments.                |
| 6    | Ch. 6 (APT/CAPM/ICAPM)                    | Factor/SDF equivalence; time-varying risk prices.              |
| 7    | Ch. 7 (PV decomposition)                  | Predictability as discount-rate dynamics.                      |
| 8    | Ch. 8 (dual puzzle)                       | Identify which moments/bounds fail.                            |
| 9    | Ch. 9 (Epstein–Zin, long-run risk)        | Preferences reshape $m_{t+1}$ .                                |
| 10   | Ch. 10–12 (heterogeneity/disasters/habit) | Compare SDF amplifiers.                                        |
| 11   | Ch. 13–14 (derivatives)                   | Replication and $Q$ -pricing as (4).                           |
| 12   | Part V (empirics)                         | Estimate/test $m_{t+1}$ ; close theory–data loop.              |

## 10. Self-check questions (quick diagnostics)

### SDF basics

1. Starting from (1), derive (2).
2. Show how (3) implies (4).
3. Interpret HJ bound (6) using Sharpe ratios.

### Factor models

1. Given  $m_{t+1} = a - b^\top f_{t+1}$ , derive linear pricing restrictions for expected returns.
2. Distinguish traded vs non-traded factors; state how moment conditions differ.

### Puzzles

1. Explain how a high equity premium and a low risk-free rate jointly constrain  $\mathbb{E}[m]$  and  $\sigma(m)$ .
2. Map each mechanism (EZ, habit, disasters, incomplete markets) to a modification in the distribution of  $m_{t+1}$ .

## Derivatives

1. State the relation between replication arguments and the existence of a risk-neutral measure.
2. Explain why option prices can be written as discounted  $Q$ -expectations.

## 11. Mini-glossary

- **SDF / pricing kernel:** random variable  $m_{t+1}$  satisfying (1).
- **State price:** price today of one unit paid in a particular state tomorrow.
- **Risk-neutral measure  $Q$ :** probability measure that prices assets via (4).
- **HJ bound:** lower bound on SDF volatility implied by observed Sharpe ratios.
- **Factor pricing:** representation of  $m_{t+1}$  in terms of a few risk factors  $f_{t+1}$ .